

Comparative Analysis of Principal Component Analysis and Autoencoders on Fashion-MNIST Dataset

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I. PCA - CLASSICAL LINEAR COMPRESSION

TABLE 1

RECONSTRUCTION ERROR FOR PCA WITH DIFFERENT COMPONENTS

COMPONENTS	RECONSTRUCTION ERROR
100	0.13346178119878344
50	0.2057961457918933
10	0.39612296417923626

How does the reconstruction error change as the number of principal components is reduced (e.g., 100, 50, 10)?

As the number of principal components is reduced from 100 to 50 and then to 10, the reconstruction error consistently increases. With 100 components, the error is approximately 0.133, which rises to about 0.206 for 50 components, and further escalates to roughly 0.396 when only 10 components are used. This trend demonstrates that a higher number of components allows for a more accurate reconstruction of the original data, as more variance is preserved, leading to a lower reconstruction error. On the other hand, reducing the number of components discards more information, resulting in a less accurate reconstruction and thus a higher error.

PCA: 100 Components

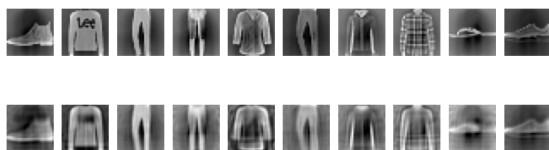


Fig. 1 Visualized Original vs Reconstructed Images for PCA with 100 Components

PCA: 50 Components

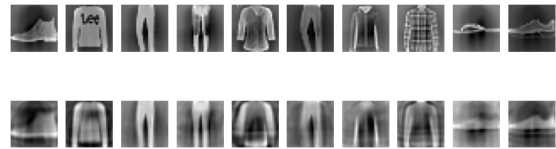


Fig. 2 Visualized Original vs Reconstructed Images for PCA with 50 Components

PCA: 10 Components

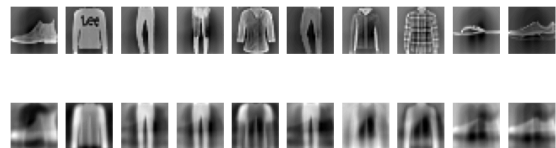


Fig. 3 Visualized Original vs Reconstructed Images for PCA with 10 Components

How does the visual quality of reconstructed images degrade as the number of components decreases?

As the number of principal components decreases, the visual quality of the reconstructed images degrades noticeably. When using 100 components, the reconstructed images retain a good level of detail and clarity, closely resembling the original items. However, the images start to show some blurring and loss of sharpness as the components are reduced to 50 with finer details becoming less distinct. This degradation becomes even more evident with only 10 components, where the reconstructed images appear significantly blurrier, more generalized, and individual features of the clothing items

are much harder to determine, indicating a substantial loss of original information.

What assumptions (e.g., linearity, orthogonality) underlie PCA? How do they affect its application in this task?

Principal Component Analysis (PCA) relies on several key assumptions that influence its effectiveness, particularly in tasks like image reconstruction (Keboola, 2022).

First, PCA assumes linearity (Gewers et al., 2021). It identifies principal components as linear combinations of the original features. This means it's best suited for data where the underlying relationships are linear. In the context of image reconstruction, if the variations in image features (pixel values) are non-linear, PCA might not fully capture the most meaningful information, leading to less optimal reconstructions. For instance, complex rotations or non-linear deformations in images are not well-represented by linear components (Singh, n.d.).

Second, PCA assumes orthogonality of the principal components. Each principal component is orthogonal (uncorrelated) to the others. This ensures that each component captures unique variance in the data (Gewers et al., 2021). For image reconstruction, this means the information captured by each successive component is independent, providing distinct layers of detail (IBM, 2025). If the underlying data structure in images were not amenable to orthogonal representation, PCA would still enforce it, potentially leading to a suboptimal dimensionality reduction.

Third, PCA is based on finding directions of maximum variance. It assumes that the directions along which the data varies the most are also the most important for retaining information (Jaadi, 2024). In image reconstruction, this means that PCA prioritizes capturing the most prominent visual variations (e.g., overall shape, major features) first. If subtle details, even if they have low variance, are crucial for human perception of image quality, PCA might discard them early when reducing dimensionality, as seen in the degradation of visual quality with fewer components (Institute of Data, 2024).

Lastly, PCA is sensitive to the scale of features and not robust against outliers (Jaadi, 2024).. While the preprocessing step of normalization (scaling pixel values to zero mean and unit variance) addresses the scale sensitivity, outliers in the dataset could disproportionately influence the direction of the principal components, potentially leading to a skewed representation of the image data and less accurate reconstructions (Iterate.ai, n.d.).

In summary, PCA's linearity and variance-maximizing assumptions mean that it excels at capturing the dominant linear patterns in image data. However, for highly complex or non-linear image variations, or if subtle low-variance details are critical, its performance in reconstruction can be limited, leading to blurrier and less detailed images as dimensionality is reduced.

II. AUTOENCODERS – NONLINEAR REPRESENTATION LEARNING

TABLE 2
TRAINING LOSS AND VALIDATION LOSS
FOR AUTOENCODER WITH DIFFERENT LATENT UNITS

LATENT UNITS	TRAINING LOSS	VALIDATION LOSS
100	0.5835	0.6065
50	0.6043	0.6221
10	0.6674	0.6835

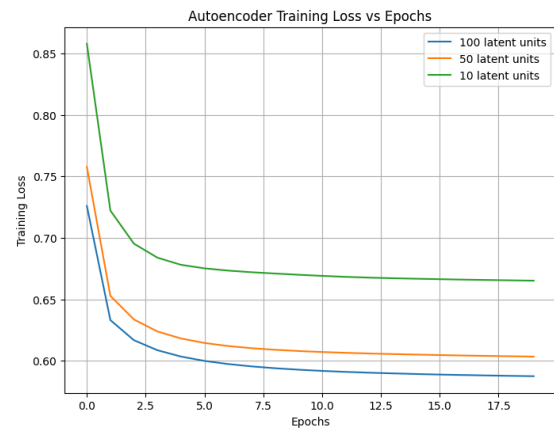


Fig. 4 Visualized Autoencoder Training Loss versus Epochs with Different Latent Units

How does the reconstruction quality change when the size of the latent space is varied (e.g., 100, 50, 10)?

As the size of the latent space in the autoencoder is reduced, the reconstruction quality degrades. This is evident by the increase in both training and validation loss. With 100 latent units, the autoencoder achieves the lowest training loss (0.5835) and validation loss (0.6065), indicating the best reconstruction quality. Decreasing the latent space to 50 units leads to a slight increase in losses (training loss: 0.6043, validation loss: 0.6221), suggesting a minor reduction in reconstruction fidelity. However, when the latent space is further reduced to just 10 units, there's a more significant rise in losses (training loss: 0.6674, validation loss: 0.6835), demonstrating a considerable degradation in the autoencoder's ability to accurately reconstruct the original images. This trend is visually consistent with the

training loss curves, where fewer latent units correspond to consistently higher loss values throughout training.

For the same compression level, how does an autoencoder's reconstruction compare with PCA?

An autoencoder typically provides a superior reconstruction quality compared to PCA, especially when dealing with complex, non-linear data like images. PCA is fundamentally a linear dimensionality reduction technique, meaning it can only capture linear relationships and project data onto orthogonal components that maximize variance. This often leads to blurrier reconstructions as more components are removed as seen in the previous answers. In contrast, autoencoders can learn intricate non-linear mappings between the input and its compressed latent representation through the use of non-linear activation functions in their hidden layers. This nonlinear capacity allows autoencoders to capture more complex and meaningful underlying structures in the data, leading to a more faithful and visually accurate reconstruction of images, even at higher compression ratios, than what a purely linear method like PCA can achieve.

What insights can be derived by visualizing the latent space (e.g., via t-SNE or PCA on the latent outputs)? Answer in one paragraph as well.

Visualizing the latent space offers crucial insights into how the model has learned to represent the input data in a compressed form. It allows us to observe the inherent structure and relationships within the dataset that the autoencoder has captured. For example, if the latent space visualization shows distinct clusters corresponding to different classes of clothing items (e.g., all "trousers" clustered together, separate from "shirts"), it indicates that the autoencoder has successfully learned meaningful, discriminative features for each class. Furthermore, the proximity of these clusters can reveal semantic similarities; if "dresses" and "skirts" are closer than "sneakers," it suggests the model understands their relatedness. This visualization helps in understanding the effectiveness of dimensionality reduction, identifying potential outliers or misclassifications, and validating whether the learned latent representation is indeed meaningful and organized in a way that reflects the underlying data distribution.

III. ADDITIONAL ANALYSIS QUESTIONS

Why is MSE (Mean Squared Error) an appropriate loss function for image reconstruction? Could alternative loss functions be used?

MSE is an appropriate loss function for image reconstruction because it measures the average squared difference between the pixel values of the original and

reconstructed images. This effectively penalizes larger errors more heavily than smaller ones which encourages the model to produce reconstructions that are pixel-wise close to the originals. For grayscale images like Fashion-MNIST, minimizing MSE directly aims to reduce discrepancies in brightness and contrast. While MSE is widely used, alternative loss functions could certainly be employed. For example, Mean Absolute Error (MAE) could be used, which is less sensitive to outliers but might lead to blurrier results. Perceptual loss functions, often derived from pre-trained deep neural networks, could also be used to evaluate reconstruction based on human perception rather than just pixel-wise differences, potentially leading to visually more appealing results even if pixel values aren't perfectly matched. Structural Similarity Index (SSIM) is another alternative that considers image structure, luminance, and contrast, which can often align better with human visual perception than simple pixel-wise errors.

What advantages do Autoencoders provide over PCA in terms of capturing nonlinear patterns?

Autoencoders offer significant advantages over PCA in capturing nonlinear patterns due to their neural network architecture. Unlike PCA, which is a linear transformation that finds orthogonal principal components, autoencoders employ multiple layers and non-linear activation functions (e.g., ReLU, sigmoid, tanh) within their encoder and decoder networks. This nonlinearity allows autoencoders to learn complex, non-linear mappings from the input data to a lower-dimensional latent representation and then back to the original space. Additionally, they can capture intricate relationships and structures in data that cannot be expressed as simple linear combinations, such as the subtle and complex variations in image features like textures, shapes, and poses. This capability enables autoencoders to create more expressive and accurate low-dimensional representations and subsequently achieve higher-quality reconstructions for non-linear data.

In what scenarios would PCA or Autoencoders be preferred for dimensionality reduction?

PCA would typically be preferred in scenarios where computational simplicity, interpretability, and linearity are key requirements, or when the dataset is relatively small. It's an excellent choice for baseline dimensionality reduction, noise reduction in linearly correlated data, and when the principal components themselves need to be understood as explicit directions of maximum variance. For instance, PCA is efficient and effective in exploratory data analysis or when dealing with datasets known to have strong linear relationships between features. On the other hand, Autoencoders would be preferred for dimensionality reduction in

scenarios involving complex, high-dimensional data with inherent nonlinear structures, such as images, audio, or text. They are particularly advantageous when the goal is to learn a highly compressed, yet rich, representation for tasks like image reconstruction, feature learning for downstream tasks, data generation, or anomaly detection, where the ability to capture intricate non-linear patterns is crucial and interpretability of individual latent dimensions is less of a concern.

IV. REFERENCE

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